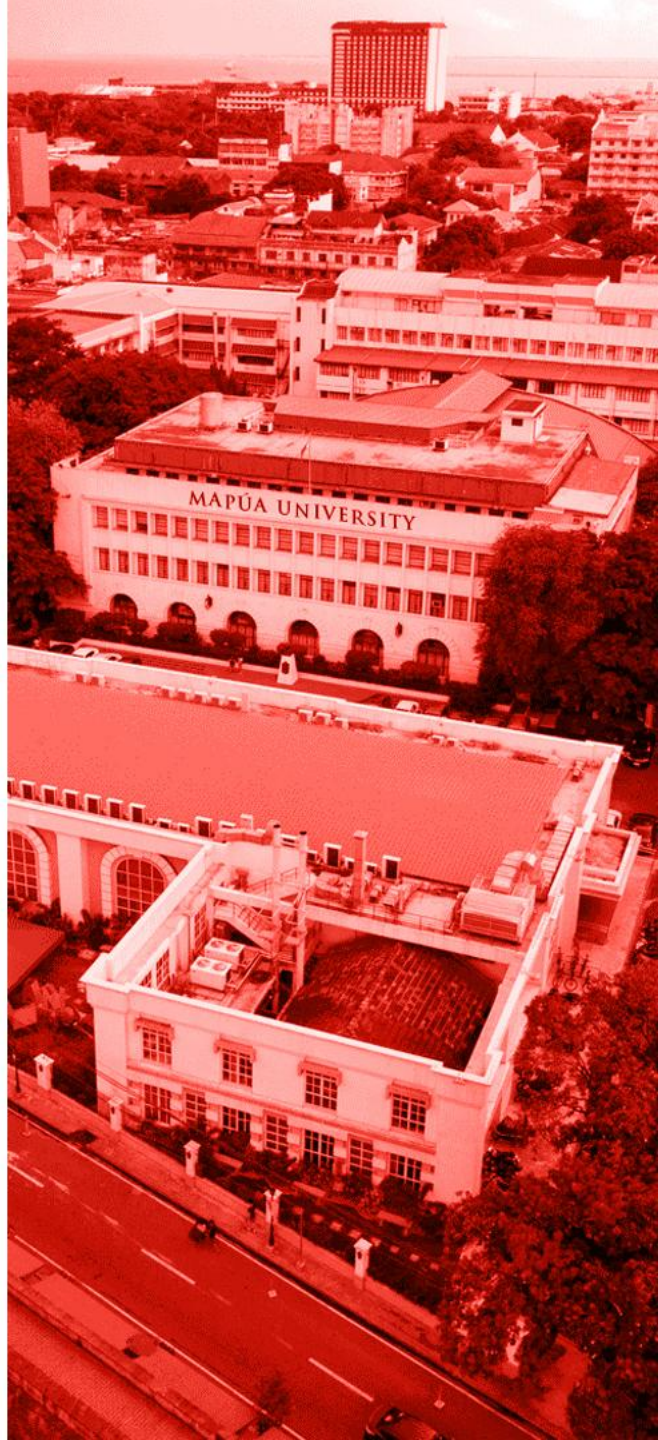
MAPÚA UNIVERSITY

SCHOOL OF ELECTRICAL, ELECTRONICS, AND COMPUTER ENGINEERING

Introduction to Hardware Description Languages and Digital Design

CPE114L-4 Introduction to HDL

John Paul T. Cruz, Ph.D.



Lesson Objectives

- Describe the role of hardware description languages in modern digital design.
- Describe the fundamentals of design abstraction in modern digital design.
- Describe the modern digital design flow based on hardware description languages.

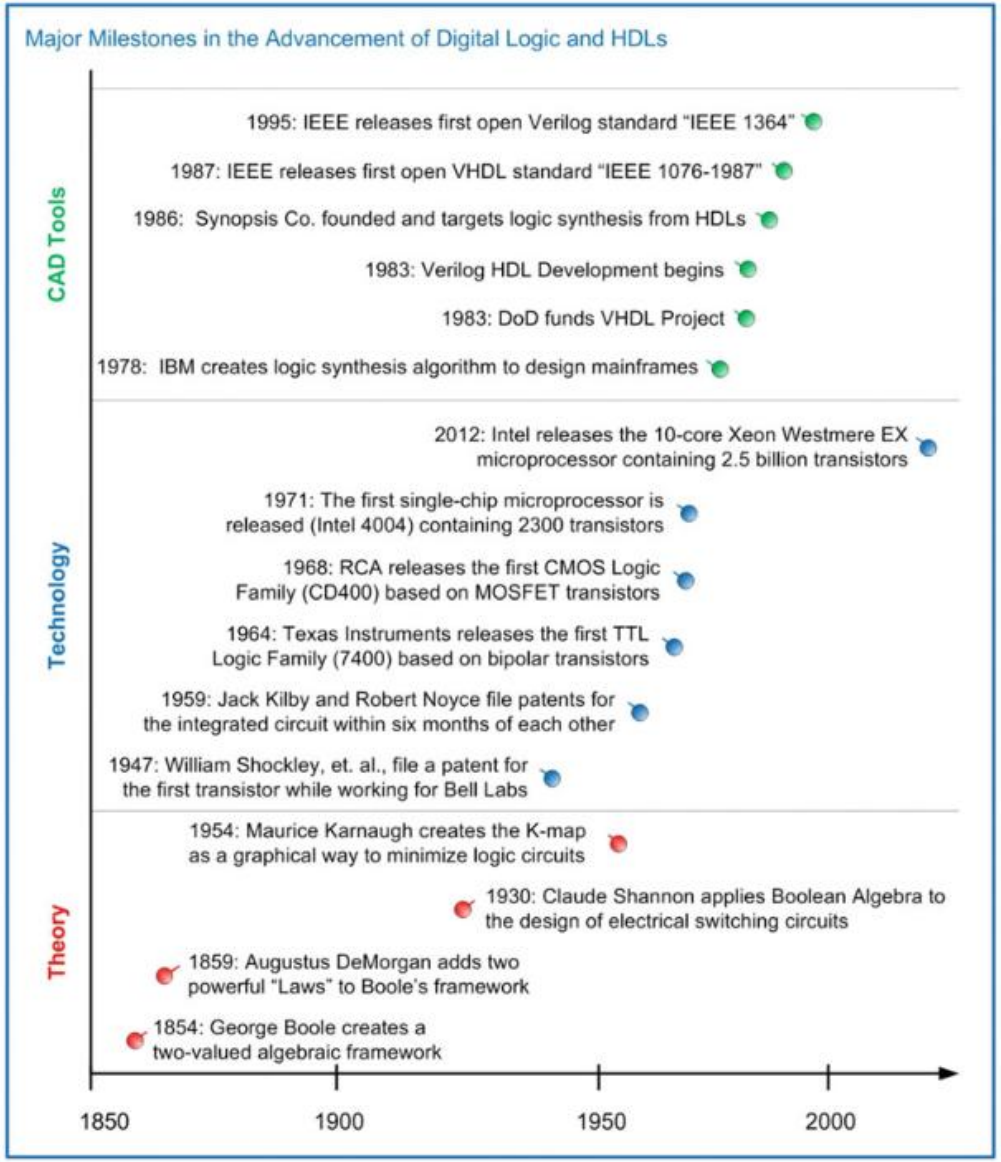
Introduction

Hardware description languages (HDLs) are used to describe digital circuitry using a text-based language. HDLs provide a means to describe large digital systems without the need for schematics, which can become impractical in very large designs. HDLs have evolved to support logic simulation at different levels of abstraction.

History

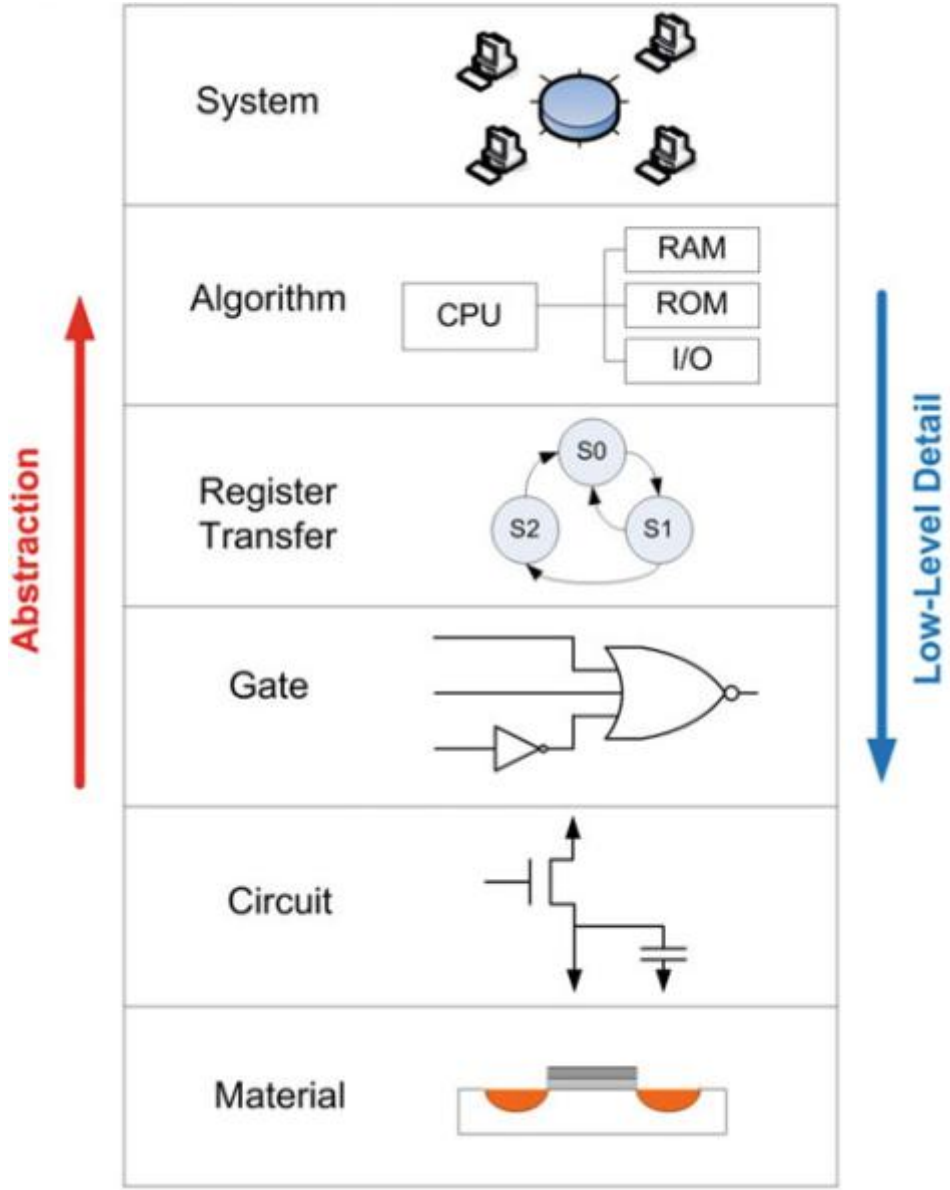
Here is a short history on the development of hardware description languages:

- The development of integrated circuits in the late 1950s spurred the growth of the semiconductor industry and led to rapid advancements in technology. As integrated circuits became more complex, there was a growing need for better ways to document and simulate their behavior.
- In 1983, the U.S. Department of Defense initiated a program to create a standardized tool for documenting digital systems, leading to the development of VHDL.
- VHDL was released in 1985 and later standardized by the IEEE. Around the same time, Verilog HDL was developed independently as a logic simulation language. It gained popularity, and Cadence Design Systems made it open to the public. The IEEE also standardized Verilog.
- The development of CAD tools for logic synthesis, particularly by Synopsys in 1986, further advanced the use of HDLs by integrating design and synthesis into a single flow.
- Initially, HDLs were used for documentation and simulation, with logic synthesis tools developed later. This has led to challenges for designers, as not all HDL-described behavior can be synthesized

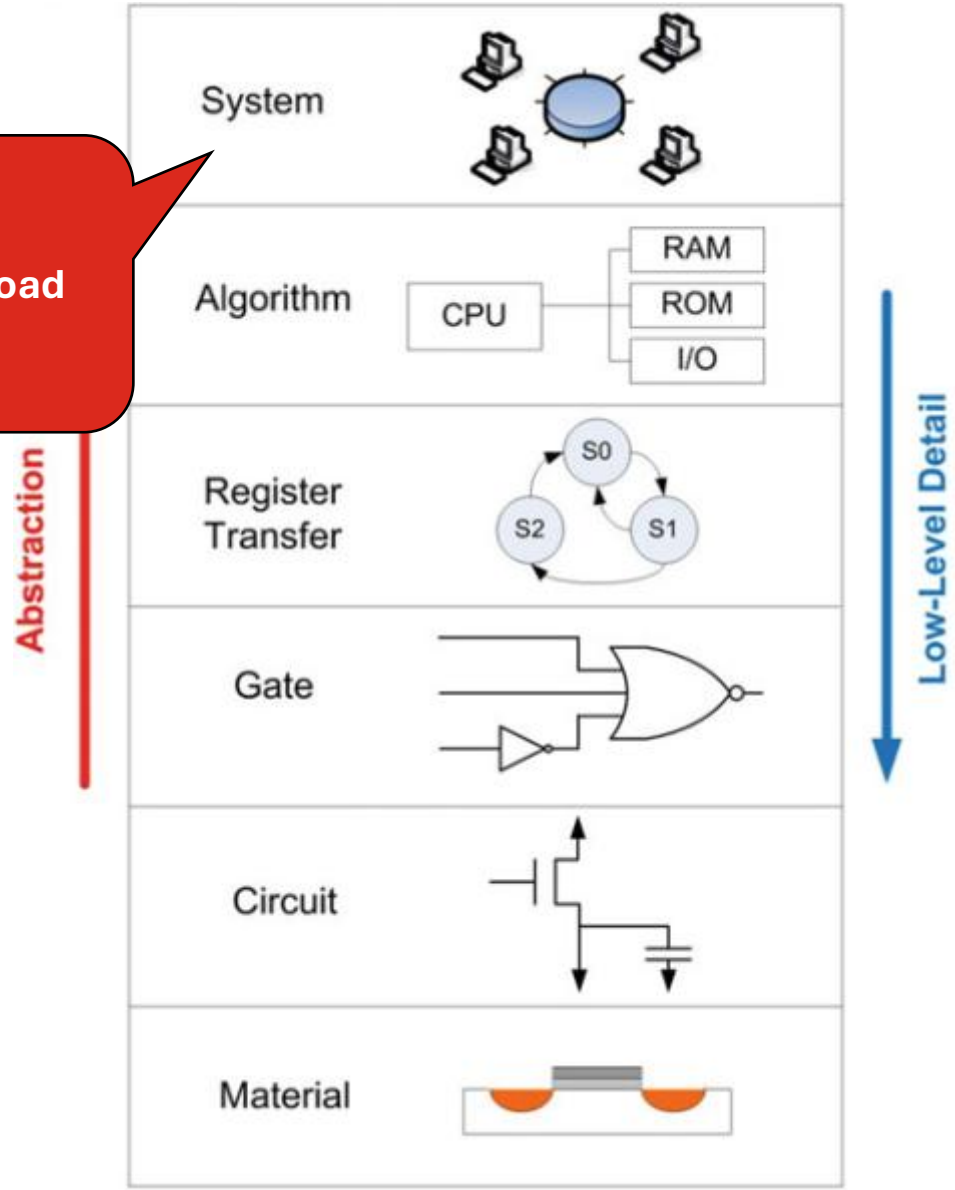

Major Milestones in the Advancement of Digital Logic and HDLs
CAD Tools
1995: IEEE releases first open Verilog standard "IEEE 1364"
1987: IEEE releases first open VHDL standard "IEEE 1076-1987"
1986: Synopsis Co. founded and targets logic synthesis from HDLs
1983: Verilog HDL Development begins
1983: DoD funds VHDL Project
1978: IBM creates logic synthesis algorithm to design mainframes
Technology
2012: Intel releases the 10-core Xeon Westmere EX microprocessor containing 2.5 billion transistors
1971: The first single-chip microprocessor is released (Intel 4004) containing 2300 transistors
1968: RCA releases the first CMOS Logic Family (CD400) based on MOSFET transistors
1964: Texas Instruments releases the first TTL Logic Family (7400) based on bipolar transistors
1959: Jack Kilby and Robert Noyce file patents for the integrated circuit within six months of each other
1947: William Shockley, et. al., file a patent for the first transistor while working for Bell Labs
Theory
1954: Maurice Karnaugh creates the K-map as a graphical way to minimize logic circuits
1930: Claude Shannon applies Boolean Algebra to the design of electrical switching circuits
1859: Augustus DeMorgan adds two powerful "Laws" to Boole's framework
1854: George Boole creates a two-valued algebraic framework
1850
1900
1950
2000

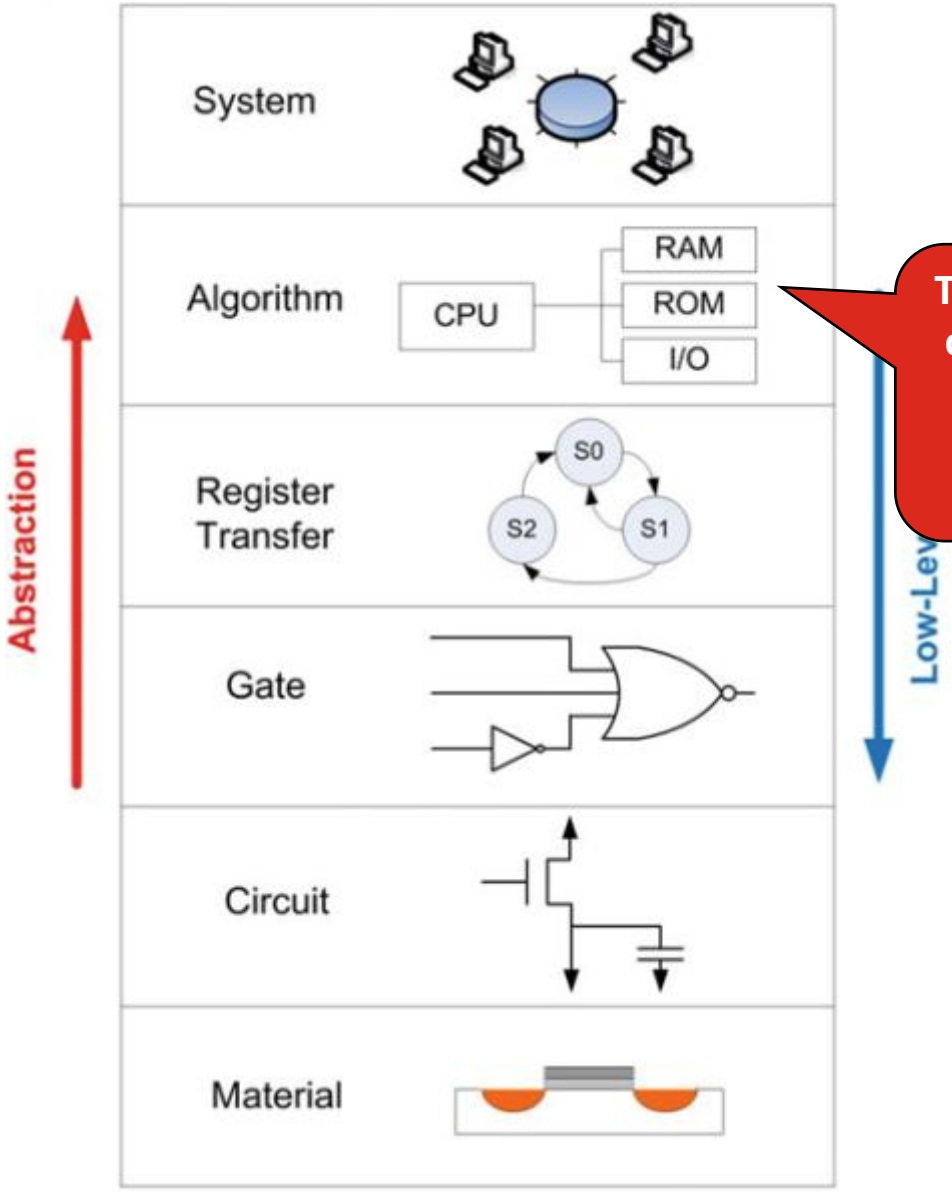
HDL Abstraction

HDLs were originally defined to be able to model behavior at multiple levels of abstraction. Abstraction is an important concept in engineering design because it allows us to specify how systems will operate without getting consumed prematurely with implementation details. By removing the details of the lower-level implementation, simulations can be conducted in reasonable amounts of time to model the higher-level functionality.


System
Algorithm
CPU
RAM
ROM
I/O
Register Transfer
S0
S1
S2
Gate
Circuit
Material
Abstraction
Low-Level Detail

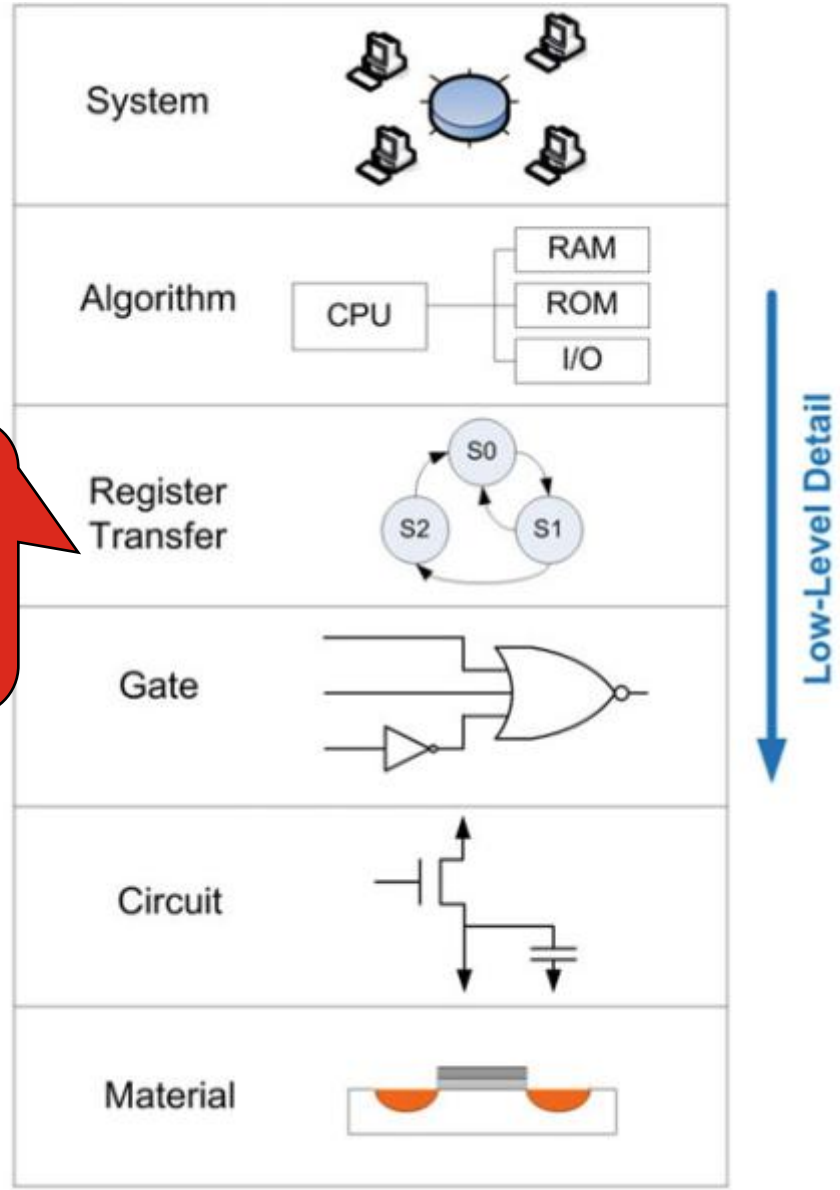
The behavior of a system is described by stating a set of broad specifications.

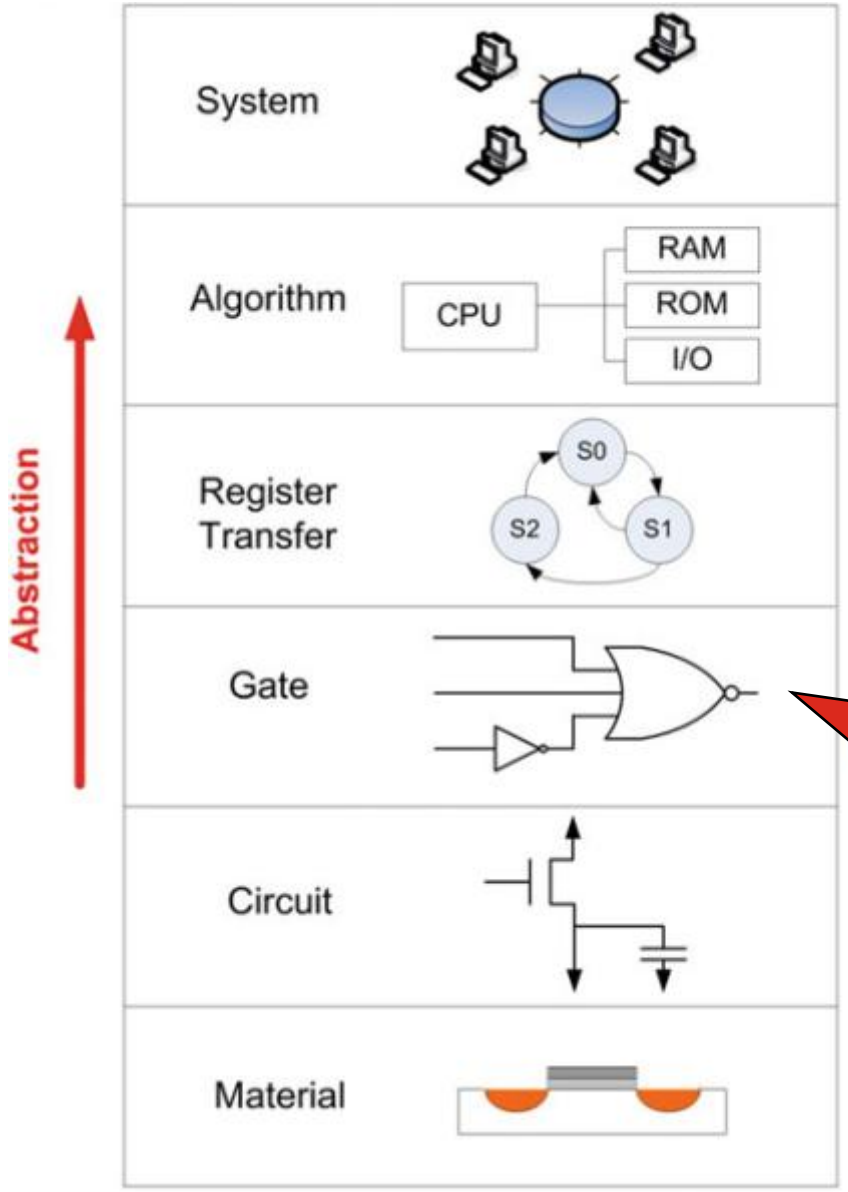




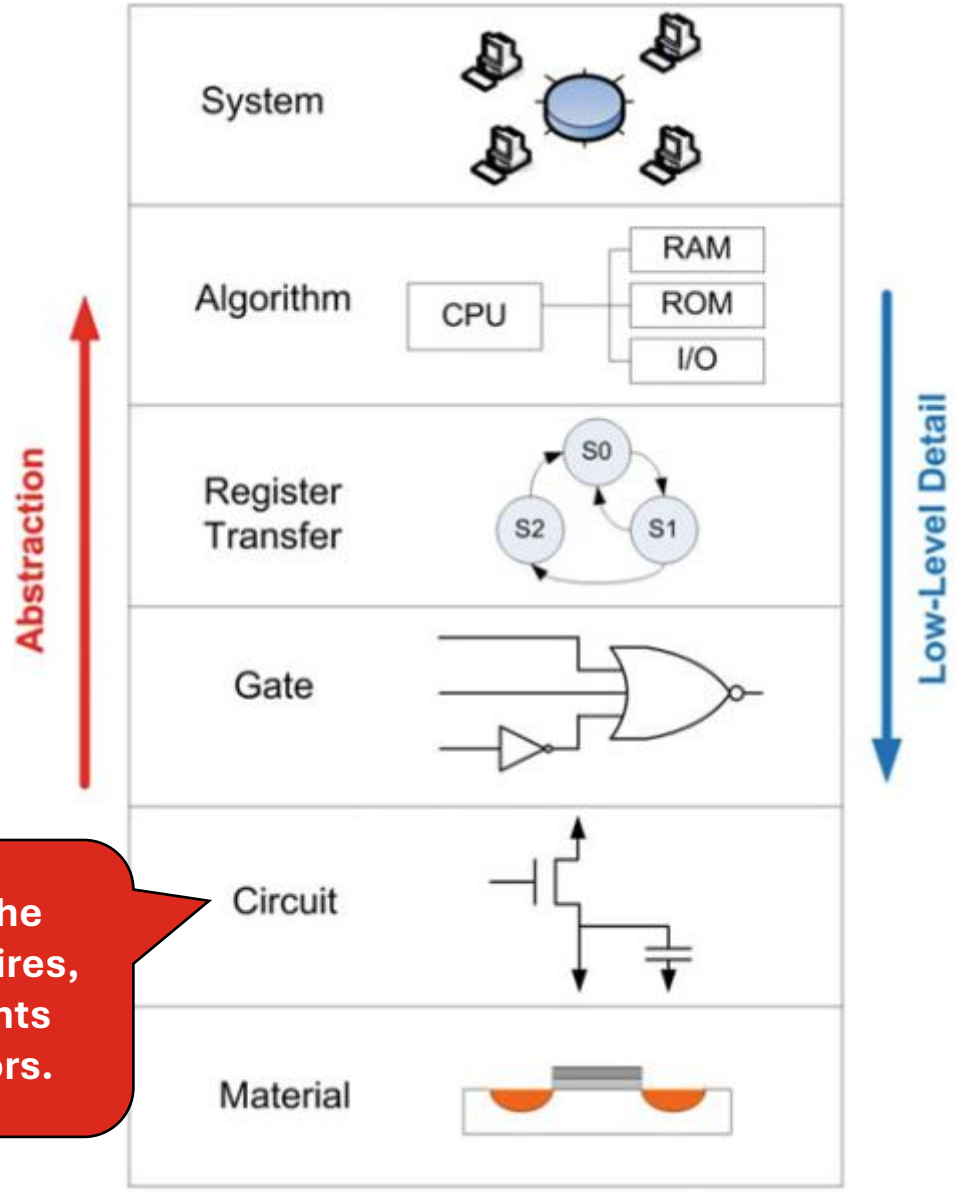
The specifications begin to be broken down into subsystems, each with an associated behavior that will accomplish a part of the primary task.

The details of how data is moved between and within subsystems are described in addition to how the data is manipulated based on system inputs.

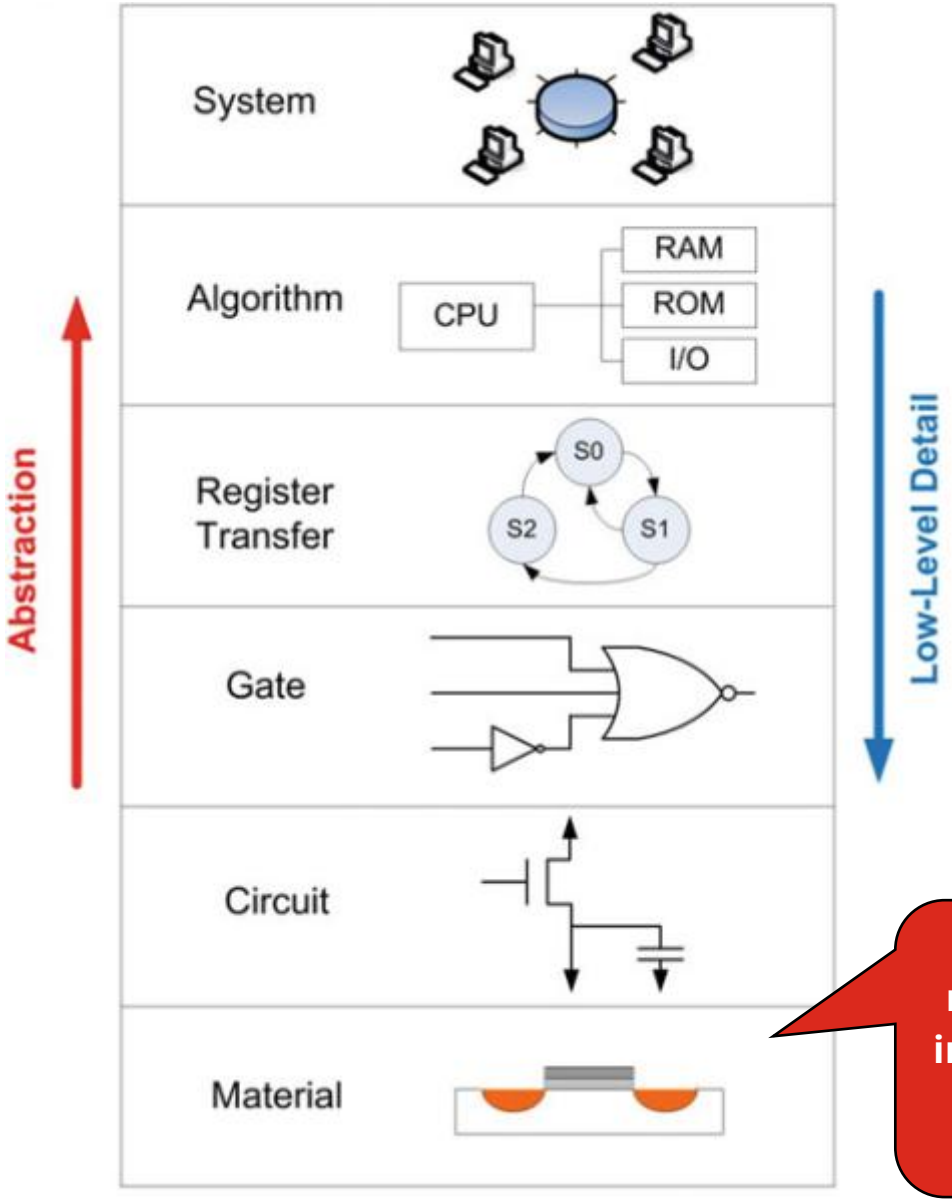




The design is a schematic that contains the basic gates, components, and connections that implements the functionalities from above levels of abstraction.



It describes the operation of the basic gates using transistors, wires, and other electrical components such as resistors and capacitors.

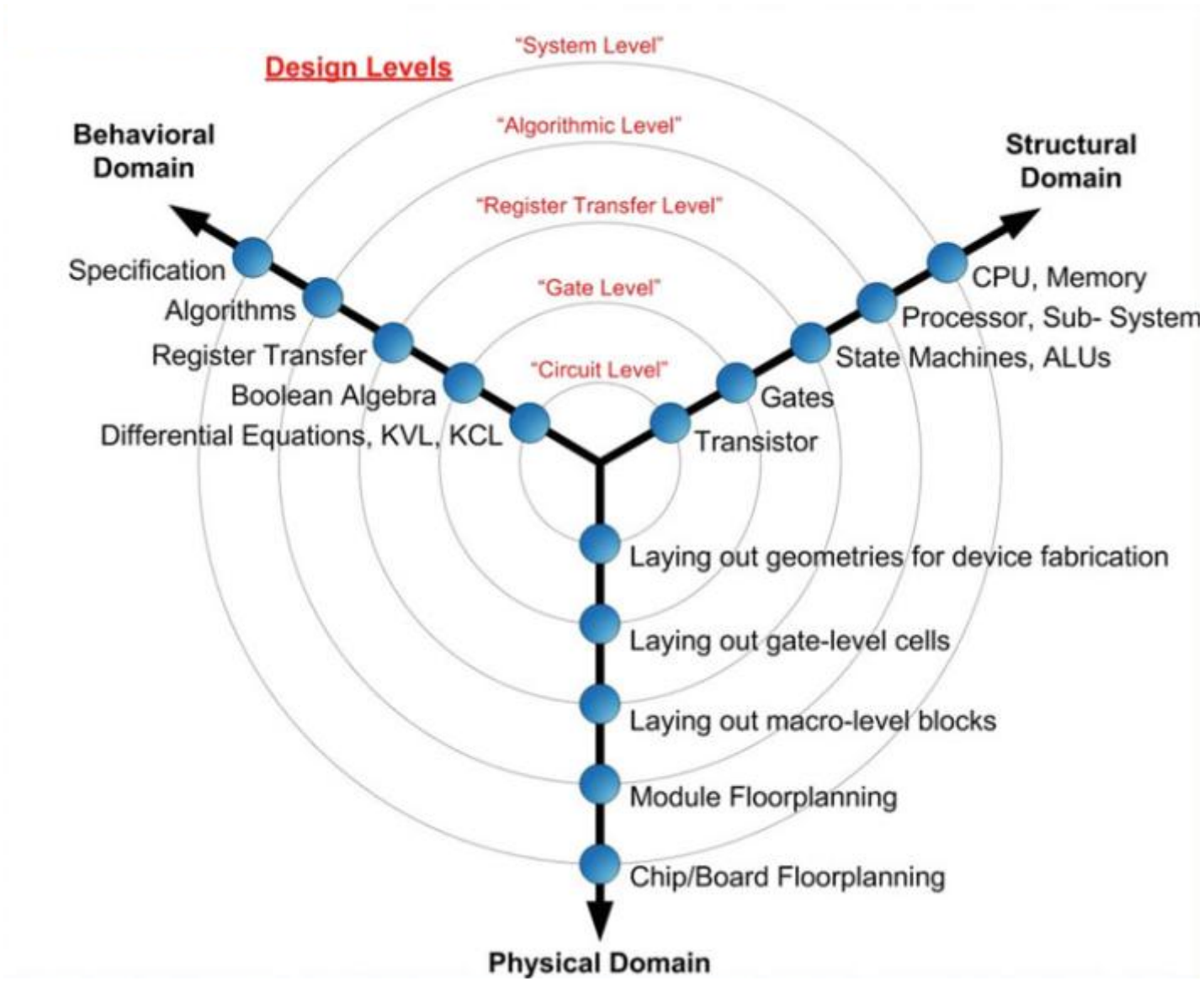


This level describes how different materials are combined and shaped in order to implement the transistors, devices, and wires from the circuit level.

HDL Abstraction

HDLs are designed to model behavior at all of these levels with the exception of the material level. While there is some capability to model circuit-level behavior such as MOSFETs as ideal switches and pull-up/pull-down resistors, HDLs are not typically used at the circuit level.

Another graphical depiction of design abstraction is known as Gajski and Kuhn's Y-chart. A Y-chart depicts abstraction across three different design domains: behavioral, structural, and physical. Each of these design domains contains levels of abstraction.


Design Levels
"System Level"
"Algorithmic Level"
"Register Transfer Level"
"Gate Level"
"Circuit Level"
Behavioral Domain
Structural Domain
Physical Domain
Specification
Algorithms
Register Transfer
Boolean Algebra
Differential Equations, KVL, KCL
CPU, Memory
Processor, Sub- System
State Machines, ALUs
Gates
Transistor
Laying out geometries for device fabrication
Laying out gate-level cells
Laying out macro-level blocks
Module Floorplanning
Chip/Board Floorplanning

The Modern Digital Design Flow

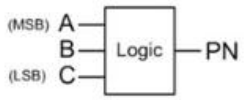
When performing a smaller design or the design of fully contained subsystems, the process can be broken down into individual steps. This process is given generically and applies to both classical and modern digital design. The distinction between classical and modern is that modern digital design uses HDLs and automated CAD tools for simulation, synthesis, place and route, and verification.

Classical Digital Design Flow

Specifications

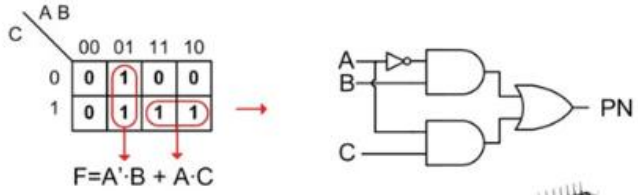
- Design a "Prime Number Detector" that takes in values from 0_{10} to 7_{10}. The circuit should be able to indicate a prime number with a delay less than 200ns.

Functional Design



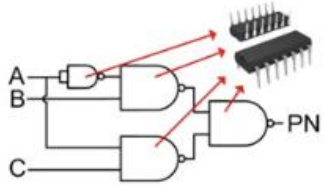
A	B	C	PN	
0	0	0	0	
0	0	1	0	
0	1	0	1	← 2
0	1	1	1	← 3
1	0	0	0	
1	0	1	1	← 5
1	1	0	0	
1	1	1	1	← 7

Synthesis



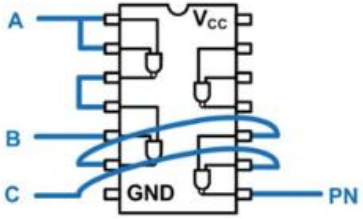
Technology Mapping

- It is decided that a 74HC logic family will be the most cost-effective technology for this design. To minimize the number of parts, the logic will be implemented with only NAND-gates.



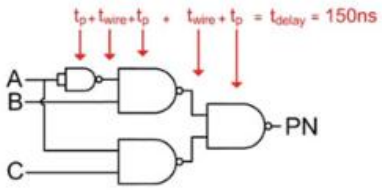
Place and Route

- The circuit to be implemented is placed in a floor plan and an estimate of the connections are made.



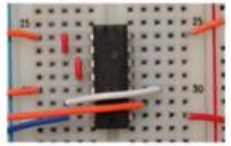
Verification

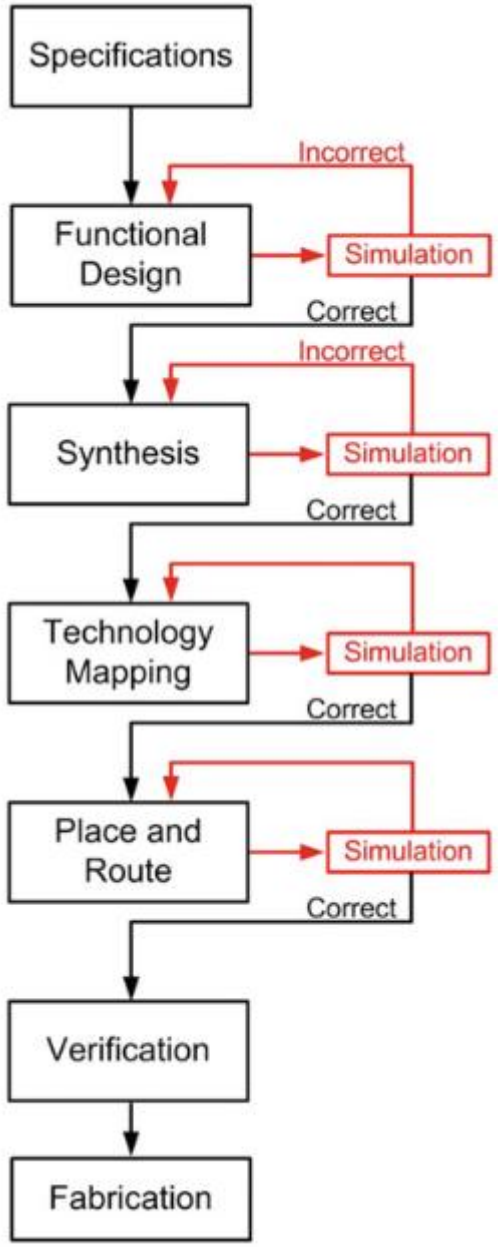
- Based on the layout, the wire delays are found. The delays of the gates are taken from the data sheet.



Fabrication

- The verified circuit is implemented in hardware.





- The initial design is in the form of an HDL behavioral description. This design is simulated to verify its proper functionality.

- After synthesis, the design is described at the gate-level. A logic simulation is used to verify that the functionality of the gate-level logic matches the functionality of the pre-synthesis behavioral description.

- After technology mapping, an estimate of the gate delays can be used in the simulation to make sure the timing requirements of the design are met.

- After place and route, an estimate of the wiring delays can be included in the simulation to make sure the timing requirements of the design are met.

- The final design is analyzed to see if it meets the original design specifications.

- Fabrication is typically in the form of an ASIC or a programmable device.

Modern Digital Design Flow



THANK YOU!